density

$$\rho(x_1, x_2, p_1, p_2).$$

A classical pure state is just a special case of a probability density, in which ρ is nonzero at only one point. But more generally, ρ will be smeared out, in which case we could call it a classical mixed state.[5] When ρ is smeared out, it means our knowledge of the system state is incomplete. The more smeared out it is, the greater our ignorance.

One thing should be completely obvious from this example: if you know the pure state for the combined two-particle system, then you know everything about each particle. In other words, a pure state for two *classical* particles implies a pure state for each of the individual particles.

But this is exactly what is *not* true in quantum mechanics when a system is entangled. The state of a composite system can be absolutely pure, but each of its constituents must be described by a mixed state.

Let's take a system composed of two parts, *A* and *B*. It could be two spins or any other composite system.

In this case, we will suppose that Alice has complete knowledge of the state of the combined system. In other words, she knows the wave function

$$\psi(a, b).$$

There is nothing missing from her knowledge of the combined system. Nevertheless, Alice is not interested in *B*. Instead, she wishes to find out as much as she can about *A* without looking at *B*. She selects an observable **L** that belongs to *A*, and that does nothing to *B* when it acts. The rule for calculating the expectation value of **L** is

$$\langle \mathbf{L} \rangle = \sum_{ab,a'b'} \Psi^*(a'b') L_{a'b',ab} \Psi(ab). \qquad (7.15)$$